Simplified Architecture: NativeAI

This diagram provides a high-level view of the system's core components and their interactions.

Diagram source code (mermaid)

```
subgraph DataSources["Data Sources"]
      DS1(["Mobile App (User Speech/Text)"])
      DS2(["WhatsApp API (Contributions)"])
      DS3(["Text Corpora (Open Data)"])
 end
subgraph Ingestion["Ingestion & Messaging Layer"]
      Q1{{"Message Queue (Kafka / RabbitMQ)"}}
      ETL1["ETL Jobs (Batch + Stream)"]
 end
subgraph Storage["Data Storage & Processing"]
      DB1[("Raw Data Lake")]
      DB2[("Cleaned & Aligned Corpus")]
 end
subgraph Training["Model Training Infrastructure"]
      T1["GPU Cluster(Distributed Training)"]
      T2["Experiment Tracking & Versioning"]
      T3{"Hyperparameter Tuning(AutoML)"}
 end
subgraph Evaluation["Model Evaluation & Feedback"]
      E1(["Automated Metrics (BLEU, CER, WER)"])
      E2{"Human-in-the-Loop Review (Teachers / Native Speakers)"}
 end
subgraph Serving["Model Serving & Deployment"]
      API["REST API Gateway (FastAPI/Docker/K8s)"]
      EDGE["Quantized Models(On-device/Offline)"]
      MON["Monitoring Service (Prometheus/Grafana)"]
 end
subgraph Clients["End Users"]
      C1(["Browser Extension"])
      C2(["Mobile SDK (Android)"])
      C3(["Integration in EdTech/Health Apps"])
 end
  DS1 -- speech/text --> Q1
  DS2 -- messages --> Q1
  DS3 -- documents --> ETL1
  Q1 -- streamed events --> ETL1
  ETL1 --> DB1
  DB1 -- cleaning, alignment, tokenization --> DB2
  DB2 --> T1
  T1 -- model checkpoints --> T2
  T1 --> T3 & E1 & API & EDGE
  T3 -- best configs --> T1
  E1 --> E2
  E2 -- feedback --> DB2
  E2 -- corrections --> T1
  API --> MON & C1 & C3
  EDGE --> MON & C2
```